

LECTURE 5 Supplement

Topics

Macro investment strategies, Carry trade

- One of the most popular macro strategies is the so-called *Carry Trade*
- In the carry trade investors seek to profit by borrowing in a country with low interest rates and investing in a country with high interest rates
- For the longest time, Japan was used as the borrowing country because of its low interest rates
- We first illustrate with an example how the Carry Trade works and then we'll use our Interactive Brokers account to set up such a trade

- We'll focus on the JPY-USD carry trade, where we borrow in JPY and invest in USD
- We break down the carry trade in three steps:
- **The borrowing leg**
 - Suppose we can borrow in JPY for one year at the JPY interest rate of 0.5%
 - JPY Principal borrowed = JPY 100 million
 - JPY Principal + interest to be repaid in one year is

$$\text{JPY } 100 \times (1 + 0.5\%) = \text{JPY } 100.5 \text{ million}$$

- **The investment leg**
 - Suppose we can invest at the US interest rate of 4.5%
 - Convert the JPY Principal to USD:

$$\text{JPY } 100 \times 0.0067 = \text{USD } 670,000$$

- Invest the USD at 4.5% for one year to obtain:

$$\text{USD } 670,000 \times (1 + 4.5\%) = \text{USD } 700,150$$

- Profit taking

- Convert the USD proceeds from the US investment at the end of the year back to JPY
- Suppose the spot USD/JPY in one year is
$$S_1^{USD/JPY} = 147.39$$
- Thus, the USD proceeds converted to JPY becomes:

$$\text{USD } 700,150 \times 147.39 = \text{JPY } 103,195,109$$

- Profit in JPY after repaying the JPY loan is:

$$\text{JPY } 103,195,109 - \text{JPY } 100.5 \text{ million} = \text{JPY } 2,695,108.5$$

- Profit in USD is:

$$\frac{\text{JPY } 2,695,108.5}{S_1^{USD/JPY}} = \text{USD } \frac{2,695,108.5}{147.39} = \text{USD } 18,285.56$$

- In Interactive Brokers Paper Trading account, you can implement the JPY-USD carry trade as follows
- Buy 100,000 units of USD.JPY at the spot USD/JPY 150.48
- Your **account cash** will show:
 - 1 $-JPY\ 15,048,000 = -100,000 \times 150.48$
 - 2 $+USD\ 100,000$
- Buy 100 units of US-T GOVT Bills Oct01'26 at 96.5% at a cost of $USD\ 96,500 = 100 \times 965$
- Your **account cash** will show:
 - 1 $-JPY\ 15,048,000$
 - 2 $+USD\ 3,500 = 100,000 - 96,500$
- Your **account securities** will show:
 - 1 $+ 100\ US-T\ GOVT\ Bills\ Oct01'26$